

SEMINAR III · WEEK 5 · FOUNDATIONS

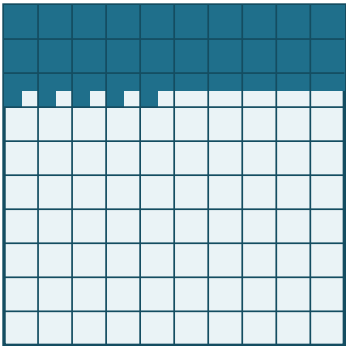
Percent

Percent means out of 100. Once you can see it as a fraction and a decimal wearing a different outfit, discounts, tax, tips, and percent change are all the same handful of moves.

SESSION A

What Percent Means, and Finding Part / Percent / Whole

PERCENT MEANS OUT OF 100



25 out of 100 squares are shaded — that's 25%. The same amount is 25/100, 0.25, or 1/4. All four are the same quantity, just written differently.

FRACTION, DECIMAL, OR PERCENT?

The representation changes. The quantity doesn't.

10% = 1/10

25% = 1/4

50% = 1/2

75% = 3/4

You'll move between all three constantly, so these benchmarks are worth knowing cold.

PERCENT SHORTCUTS WORTH KNOWING

You can find most percentages in your head by combining a few benchmarks: 1%, 5%, 10%, 20%, 25%, 50%, 75%, 100%.

15% of a number = 10% of it, plus 5% of it (half of the 10%).

WHICH PERCENT PROBLEM IS THIS?

Every percent problem is one of three types. Before calculating, name which one you have.

Type	You know...	You're finding...
Find the part	the percent and the whole	an amount
Find the percent	the part and the whole	a percentage
Find the whole	the part and the percent	the total

FINDING A PERCENT OF A NUMBER

30% of 80: write 30% as 0.30, then multiply — $0.30 \times 80 =$
24.

FINDING WHAT PERCENT ONE NUMBER IS OF ANOTHER

Divide the part by the whole, then convert to a percent.

18 out of 24 correct: $18 \div 24 = 0.75 = \mathbf{75\%}$.

FINDING THE WHOLE

Try benchmark reasoning first. Otherwise, divide the part by the decimal percent.

15 is 20% of what number? $15 \div 0.20 = \mathbf{75}$. Check: 20% of 75 is 15. ✓

BEFORE WE MOVE ON

Part, percent, or whole? Identify it — then solve.

SESSION B

Percent in Money, and Percent Change

DISCOUNTS AND TAX: LABEL WHAT YOU'RE FINDING

An \$80 jacket, 25% off: $\text{discount} = 0.25 \times 80 = \20 . Sale price = $80 - 20 = \mathbf{\$60}$.

A \$50 purchase, 8% tax: $\text{tax} = 0.08 \times 50 = \4 . Final cost = $50 + 4 = \mathbf{\$54}$.

A discount reduces the price. Tax increases it. Keep the amount and the final price straight — they're never the same number.

TIPS

You don't need exact restaurant-etiquette rules — you need a fast, useful estimate. Find 10% by moving the decimal point one place, double it for 20%, or add half of the 10% for 15%.

PERCENT INCREASE AND DECREASE

original → **new** → **difference** → **percent change**

A price rises from 40 to 50: change = 10. $10 \div 40 = 0.25 =$
25% increase.

Percent change always divides by the **original** value — using the new value instead is a common trap.

DOES MY ANSWER MAKE SENSE?

- Is the part smaller than the whole when the percent is under 100%?
- Should a discount decrease the price? Should tax increase it?
- Did I use the *original* value for percent change?
- Did I answer with the percent amount, or the final amount — whichever was actually asked for?

BEFORE WE MOVE ON

One discount or tax problem: find both the percent amount and the final total.

Then: mixed practice, error analysis, Week 5 check, reflection.

TRANSITION TO WEEK 6

Ratios, Rates & Proportions

Keep using fractions, decimals, percent, division, estimation, and reasonableness. The big Week 6 question: what two quantities am I comparing?